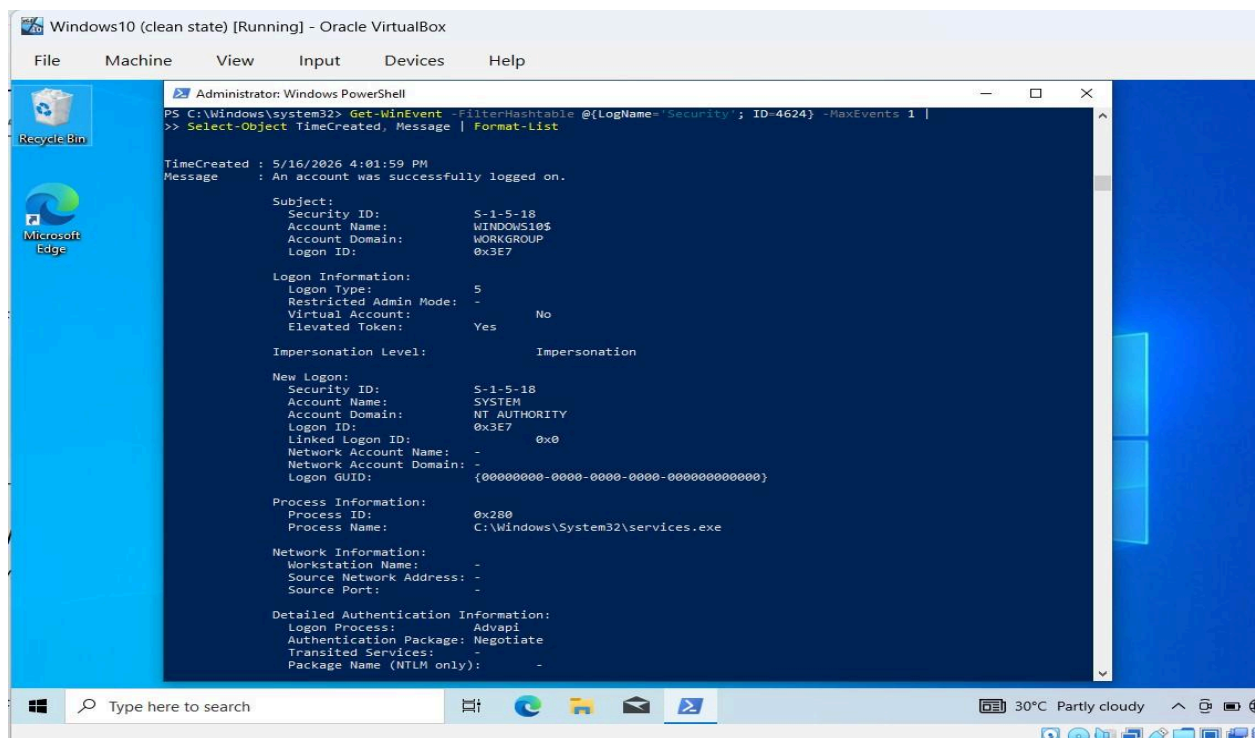
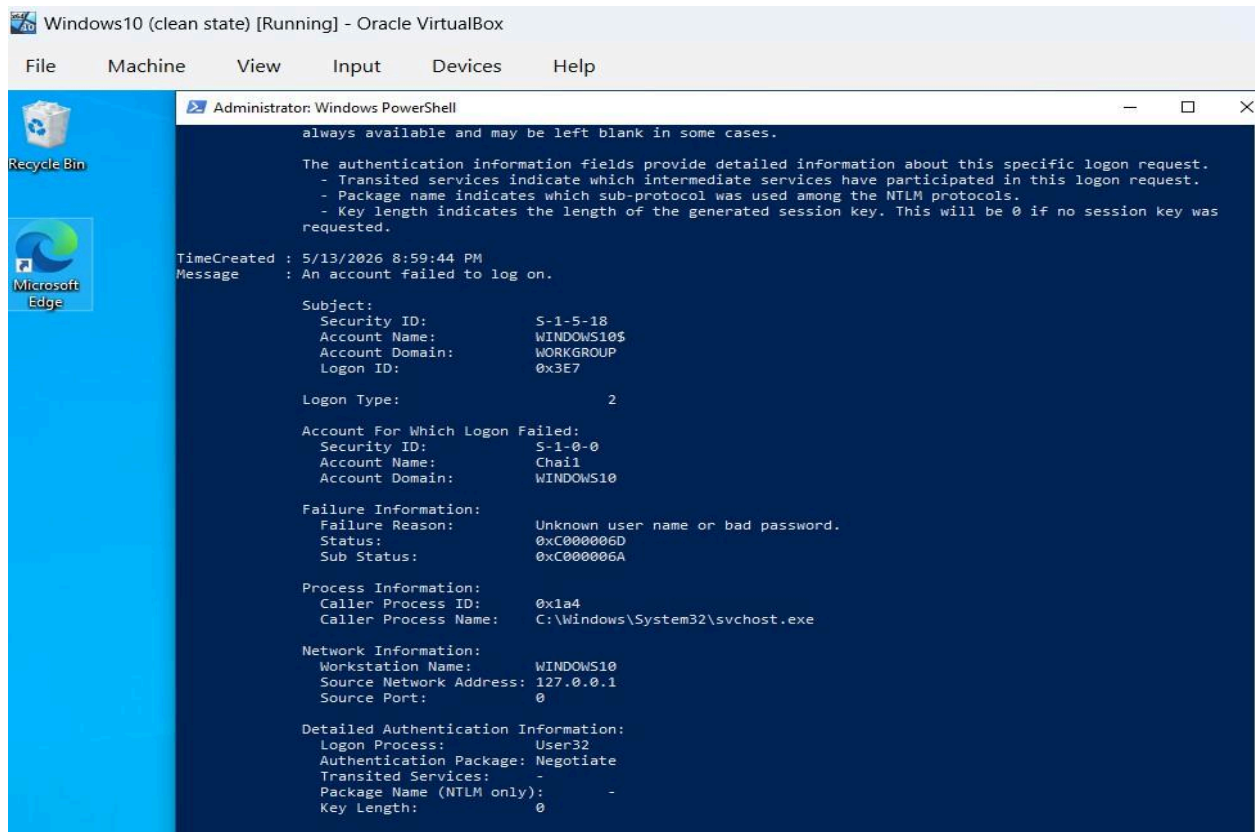


Analyze Security Log for Authentication

Deliverable: Screenshot showing failed logon details with elds highlighted



Install and Configure Sysmon
Screenshot showing Sysmon service running

```
PS C:\Windows\system32> Get-Service *sysmon*

Status      Name             DisplayName
-----
Running     Sysmon64         Sysmon64

PS C:\Windows\system32>
```

Hunt for Encoded PowerShell
Deliverable: Screenshot showing the encoded PowerShell command in Sysmon logs

```
Select Administrator: Windows PowerShell

The '<' operator is reserved for future use.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : RedirectionNotSupported

PS C:\Windows\system32> powershell -EncodedCommand VwByAGkAdAB1AC0ASABvAHMAAdAAgACcARwByAGEAeQBtAGUAbgB0AGkAbgB1AGwAJwA=
GraySentinel
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{
>> LogName='Microsoft-Windows-Sysmon/Operational'
>> ID=1
>> } |
>> Where-Object { $_.Message -like "**powershell*" } |
>> Select-Object -First 5 TimeCreated, Message | Format-List

TimeCreated : 5/16/2026 4:32:56 PM
Message      : Process Create:
               RuleName: -
               UtcTime: 2026-05-16 11:02:56.043
               ProcessGuid: {694c19db-4ee0-6a08-0201-000000000900}
               ProcessId: 4836
               Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
               FileVersion: 10.0.19041.3636 (WinBuild.160101.0800)
               Description: Windows PowerShell
               Product: Microsoft® Windows® Operating System
               Company: Microsoft Corporation
               OriginalFileName: PowerShell.EXE
               CommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -EncodedCommand
               VwByAGkAdAB1AC0ASABvAHMAAdAAgACcARwByAGEAeQBtAGUAbgB0AGkAbgB1AGwAJwA=
               CurrentDirectory: C:\Windows\system32\
               User: WINDOWS10\Chail
               LogonGuid: {694c19db-4429-6a08-f2c4-020000000000}
               LogonId: 0x2C4F2
               TerminalSessionId: 1
               IntegrityLevel: High
               Hashes: MD5=6726185B70B5ADF05E8A1A1DF82EBF30, SHA256=64DD55E1C2373DEED25C2776F553C632E58C45E56A0E4639DFD54
               EE97EAB9C19, IMPHASH=E3007C8E0098D06ABF617EEE6F0C5ABD
               ParentProcessGuid: {694c19db-470d-6a08-ab00-000000000900}
               ParentProcessId: 4712
               ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
               ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
               ParentUser: WINDOWS10\Chail

PS C:\Windows\system32>
```

Timeline Reconstruction
Timeline table with at least 10 events from the attack sequence

```
PS C:\Windows\system32> $Events |
>> Sort-Object TimeCreated |
>> Select-Object TimeCreated, Id, LogName |
>> Format-Table -AutoSize
```

TimeCreated	Id	LogName
5/18/2026 8:43:21 PM	4624	Security
5/18/2026 8:44:44 PM	4624	Security
5/18/2026 8:44:57 PM	4624	Security
5/18/2026 8:46:15 PM	4624	Security
5/18/2026 8:46:15 PM	4624	Security
5/18/2026 8:46:18 PM	4624	Security
5/18/2026 8:47:41 PM	4624	Security
5/18/2026 8:48:11 PM	4624	Security
5/18/2026 8:51:53 PM	4624	Security
5/18/2026 8:52:47 PM	4624	Security
5/18/2026 8:52:47 PM	4624	Security
5/18/2026 8:56:25 PM	4624	Security
5/18/2026 8:58:02 PM	4624	Security
5/18/2026 8:58:06 PM	4624	Security
5/18/2026 9:04:58 PM	4624	Security

Create Sysmon Event Collector Script

The script file and execution screenshot showing output

```
PS C:\Windows\system32> notepad C:\Reports\sysmon_collector.ps1
PS C:\Windows\system32> cd C:\Reports
PS C:\Reports> .\sysmon_collector.ps1 -Hours 1
.\sysmon_collector.ps1 : File C:\Reports\sysmon_collector.ps1 cannot be loaded because running scripts is disabled on
this system. For more information, see about_Execution_Policies at https://go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:1
+ .\sysmon_collector.ps1 -Hours 1
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess
PS C:\Reports> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): y
PS C:\Reports> .\sysmon_collector.ps1 -Hours 1
[+] Collecting Sysmon events from last 1 hours...
[+] Collected 20 events
[+] Report saved to C:\Reports\sysmon_events_20260518_211548.csv
PS C:\Reports>
```

